

Developer Setup Guide

A simple guide to clone, configure and run the CSV processor across local, kops and EKS environments.

Repository Structure

Clone once — everything is inside a single monorepo:

```
git clone https://github.com/asif-ahmedb/csv-processor
cd csv-processor
```

The repo contains three modules:

```
csv-processor/
├── csv-processor-app/           Flask app, Docker image, tests
├── csv-processor-infrastructure/ AWS cluster – kops or Terraform/EKS
└── csv-processor-k8s-assets/    Helm chart, Ansible, Minikube scripts
```

Prerequisites

All paths require Git 2.x and Docker Desktop 4.x.

Local / Minikube also needs: - Minikube - kubectl - Helm 3 - Python 3.10+

AWS paths also need: - AWS CLI v2 - kubectl - Helm 3 - Git Bash (Windows — needed for envsubst) - kops (kops path only) - Terraform 1.5+ (EKS path only)

AWS IAM permissions required: EC2, VPC, ELB, S3, IAM, Route 53 (kops), EKS (Terraform).

Verify AWS credentials:

```
aws sts get-caller-identity
```

Path 1 — Local

Run with Docker:

```
cd csv-processor-app
docker build -t csv-processor:local .
docker run --rm -p 8081:8081 -e S3_BUCKET="" -e AWS_REGION=us-east-1 csv-processor:local
```

Open <http://localhost:8081>

Run tests:

```
cd csv-processor-app/web
pip install -r requirements.txt
pytest tests/ -v
```

Deploy on Minikube:

```
cd csv-processor-k8s-assets/minikube
./deploy.sh
kubectl port-forward -n csv-processor svc/csv-processor 8080:80
```

Open <http://localhost:8080>

Path 2 — AWS with kops

Configure:

```
cd csv-processor-infrastructure/kops
cp settings.env.example settings.env
```

Edit settings.env — set at minimum:

```
AWS_ACCOUNT_ID=<YOUR_ACCOUNT_ID>
AWS_REGION=us-east-1
```

Bootstrap cluster (~15 min):

```
chmod +x scripts/*.sh
./scripts/bootstrap.sh
```

Deploy application:

```
cd ../../csv-processor-k8s-assets
source ../csv-processor-infrastructure/kops/settings.env
helm upgrade --install csv-processor ./helm/csv-processor \
  -f ./helm/csv-processor/values-kops.yaml \
  --namespace csv-processor --create-namespace \
  --set config.s3Bucket="${PROCESSED_BUCKET}" \
  --set-string aws.accountId="${AWS_ACCOUNT_ID}"
```

Access: Wait for EXTERNAL-IP, then open <http://<EXTERNAL-IP>>

```
kubectl get svc -n csv-processor csv-processor -w
```

Teardown:

```
cd csv-processor-infrastructure/kops
./scripts/teardown.sh --yes
```

Path 3 — AWS with Terraform / EKS

Provision infrastructure:

```
cd csv-processor-infrastructure/terraform
cp terraform.tfvars.example terraform.tfvars
terraform init && terraform plan && terraform apply
```

Configure kubectl:

```
aws eks update-kubeconfig \
  --name $(terraform output -raw eks_cluster_name) \
  --region $(terraform output -raw aws_region)
```

Deploy application:

```
cd ../../csv-processor-k8s-assets
helm upgrade --install csv-processor ./helm/csv-processor \
  -f ./helm/csv-processor/values-eks.yaml \
  --namespace csv-processor --create-namespace \
  --set config.s3Bucket=$(terraform -chdir=../csv-processor-infrastructure/terraform output -raw s3_bucket) \
  --set aws.irsRoleArn=$(terraform -chdir=../csv-processor-infrastructure/terraform output -raw aws_irs_role_arn)
```

Access:

```
kubectl port-forward -n csv-processor svc/csv-processor 8080:80
```

Open <http://localhost:8080>

Teardown:

```
cd csv-processor-infrastructure/terraform
terraform destroy
```

CI/CD

On every push to main that changes csv-processor-app/, GitHub Actions: 1. Runs pytest 2. Builds the Docker image 3. Pushes to Docker Hub

Add these secrets in csv-processor → Settings → Secrets and variables → Actions: - DOCKERHUB_USERNAME - DOCKERHUB_TOKEN

Reference Documents

kops provisioning:

https://github.com/asif-ahmedb/csv-processor/blob/main/csv-processor-infrastructure/kops/KOPS_QUICK

Application deployment:

https://github.com/asif-ahmedb/csv-processor/blob/main/csv-processor-k8s-assets/DEPLOY_QUICK_RE

Minikube:

https://github.com/asif-ahmedb/csv-processor/blob/main/csv-processor-k8s-assets/minikube/MINIKUBE_